Machine Learning for Raman and SERS Spectral Classification:

A Summer Research Project Report

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Abstract

The classification of chemical species using Raman and Surface-Enhanced Raman Spectroscopy (SERS) is often hindered by spectral similarity and variability, particularly in SERS where different substrates can produce inconsistent signals. This project presents a systematic investigation into machine learning methodologies to overcome these challenges, progressing from rule-based algorithms to advanced deep learning architectures. Beginning with the implementation of the Characteristic Peak Similarity (CaPSim) method, initial results on standard Raman spectra showed limited Top-1 classification accuracy (30%), which was subsequently improved to 100% by integrating a k-Nearest Neighbors (kNN) classifier. Motivated by this feature-based success, the project transitioned to Siamese Networks to learn optimal spectral representations directly from the data. This approach yielded perfect accuracy for single-component standard Raman spectra and >98% accuracy for challenging SERS data by learning to distinguish unique chemical-substrate pairs. Furthermore, a two-stage Siamese Network and Multi-Layer Perceptron (MLP) model successfully classified two-component mixtures from their convoluted spectra, achieving a weighted F1-score of 0.91 on real experimental data after being trained on synthetic mixtures. A final clustering analysis quantitatively confirmed that the learned Siamese embeddings provide a geometrically superior representation of the data compared to the raw spectra, especially for SERS. This work conclusively demonstrates that deep learning models, particularly Siamese Networks, are a powerful and robust tool for the accurate classification of complex and variable spectroscopic data.

1 Introduction

This report provides a comprhensive summary of the research and work conducted during the summer work term on the classification of Raman and Surface-Enhanced Raman Spectroscopy (SERS) signals using machine learning (ML) techniques. The primary objective of this project was to explore and develop robust methods for identifying chemical species from their vibrational spectra, addressing common challenges such as spectral variability, similarity in spectral fingerprint and the analysis of complex mixtures.

This document is intended for an audience with a strong background in chemistry but limited familiarity with ML. Therefore, key ML concepts are explained.

This report details the project's evolution, from initial exploratory analyses and the implementation of established methods to the implementation of advanced neural network architectures and a deeper investigation into the learned data representations.

2 Project Inception & Initial Analyses

2.1 Inspiration: The 2023 Halas Paper & CaPSim

The project direction was initially inspired by a 2023 paper by the Halas group, introducing a novel method called **Characteristic Peak Similarity (CaPSim)** [1]. This method addresses a significant challenge in SERS: the high degree of spectral variability that arises from different SERS substrates, which can make it difficult to use standard Raman libraries for visual identification.

The CaPSim method, as detailed in [1], draws an analogy to facial recognition, where models learn a person's most representative features to ensure identification despite variations in new images. Similarly, instead of comparing entire spectra, it focuses on identifying and comparing a set of *characteristic peaks* (CPs) that are most representative of a given analyte. According to [1], this approach makes the classification of SERS signal robust to nuisance variables like peak shifts and intensity fluctuations, which are common in SERS but do not reflect the chemical identity of the analyte.

2.2 Datasets used

2.2.1 Standard Raman Single Spectrum Dataset

The following dataset, shown in Table 1, was divided into the reference (sometimes referred to as the training set) and query set (testing set) in a 50/50 split, unless otherwise explained. This dataset was used for single spectrum classification.

2.2.2 Standard Raman Mixtures Dataset

Table 2 shows the dataset of two chemical mixtures used for two-component Raman classification.

Chemical	**# of Examples**
1,9-nonanedithiol	27
1-dodecanethiol	30
1-undecanethiol	26
6-mercapto-1-hexanol	27
Benzene	27
Benzenethiol	27
DMMP	27
Ethanol	135
Methanol	157
n,n-dimethylformamide	28
Pyridine	27
Tris(2-ethylhexyl) phosphate	27

Table 1: The dataset of standard Raman single spectra used as the reference (training) and query (testing) sets.

Chemical	**# of Examples**
1-dodecanethiol	243
6-mercapto-1-hexanol	108
Benzene	193
Benzenethiol	72
Ethanol	121
Methanol	243
n,n-dimethylformamide	72
Pyridine	108

Table 2: Support per chemical in the *mixtures* dataset (appearance count across both mixture components). Total mixtures: 580 (1160 total label-occurrences).

2.2.3 SERS Dataset

The following dataset, Table 3, of single spectra SERS data was used for the subsequent SERS signal classification.

Chemical	# of Examples
4np_AgNP	20
4np_PICO	20
4np_pSERS	20
Benzenethiol_Ag	20
Benzenethiol_Au	20
Benzenthiol_PICO	20
Benzenethiol_pSERS	20
n,n-dimethylformamide_AuNP	183
Pyridine_AgNP	20
Pyridine_AuNP	20
Pyridine_PICO	20
Pyridine_pSERS	20

Table 3: Support per class in this dataset (counts per class).

2.3 Exploratory Data Analysis of Standard Raman Spectra

We begin with single spectrum classification, utilizing the dataset described in Section 2.2.1. Prior to delving right into model building, the initial step was exploratory analysis of the data.

The goal was to understand the inherent structure and separability of the data before applying any classification models. This was accomplished through two primary techniques:

1. **Principal Component Analysis (PCA)**: PCA is a dimensionality reduction technique that finds orthogonal directions (Principal Components (PCs)) capturing the most variance in the data (PC1, PC2, etc.). In our case, the Raman spectrum are high-dimensional vectors we can project onto the first two PCs to obtain a 2D scatter plot. This provides an initial, qualitative view of structure or separation among samples; how well the classes can be distinguished based on the first two components that provide the most variance.

2. **Centroid Correlation**: To obtain a more quantitative measure of the similarity between the different classes/chemicals, I computed the correlation between the class centroids. A centroid is simply the average spectrum for a given chemical. A high correlation between the centroids of two different classes indicates that their spectral signatures are very similar, which could make them difficult to distinguish. This analysis helped to identify potential challenges in the classification task.

In this dataset, the first two principal components capture 54.77% of the total variance (PC1: 33.42%, PC2: 21.35%). PC1 and PC2 are primarily driven by the following peaks: $1035\,\mathrm{cm}^{-1}$, $994\,\mathrm{cm}^{-1}$, $885\,\mathrm{cm}^{-1}$, and $656\,\mathrm{cm}^{-1}$. These wavenumber regions therefore contribute disproportionately to the observed spread along PC1/PC2 and to the distinction between classes in the score plot. As seen in Table 4 below.

Principal Component	**Explained Variance (%)**	**Features (cm^{-1})**
PC1	33.42	1035, 885
PC2	21.35	885, 994, 656

Table 4: Explained variance and corresponding features from PC1 and PC2.

Consistent with Figure 1 below, most chemicals appear well separated in this space, while the structurally related alkanethiols and 6-mercapto-1-hexanol cluster closely, reflecting their highly similar spectral signatures.

The tight, almost point-like appearance of each class in the PCA plot, Figure 10a, further suggests very low within-class variability within the PC1/PC2 plane after preprocessing.

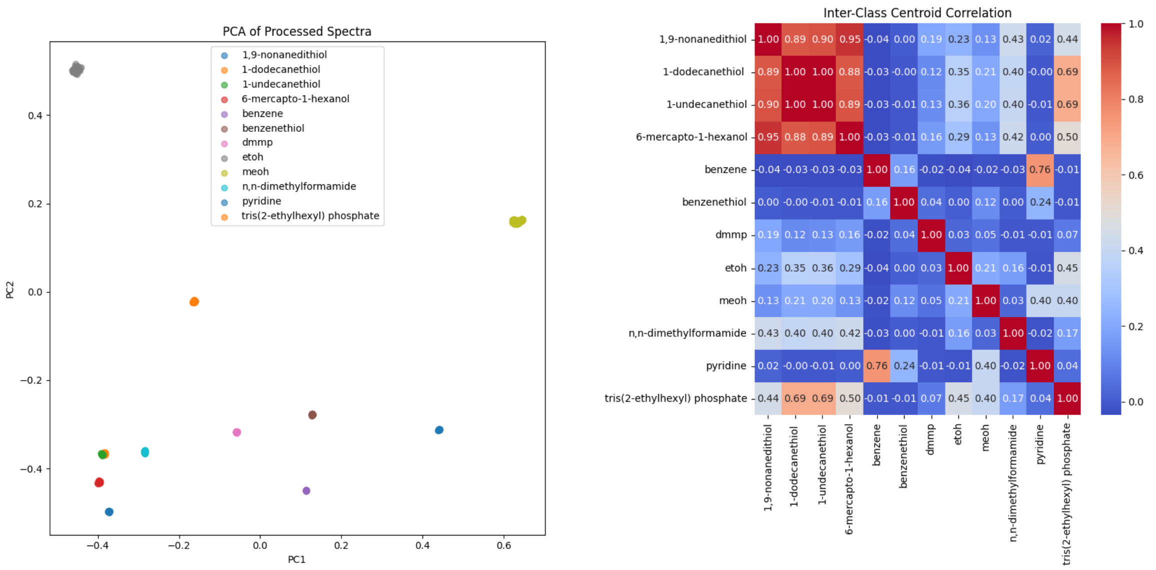
3 From Similarity Metrics to Classification Models

3.1 Implementing CaPSim

The CaPSim workflow can be conceptually divided into two main stages: the creation of a reference library of characteristic spectral features, and the subsequent use of the library to classify unknown query spectra via similarity scoring.

3.1.1 Preprocessing[†]

[†]This preprocessing step is also done for all subsequent methods used throughout the project.



(a) PCA (PC1 vs. PC2) of processed spectra. Classes occupy distinct regions with very tight—nearly point-like—within-class spread in this projection.

(b) Inter-class centroid (mean-spectrum) correlation. Warmer colors indicate greater similarity; the three alkanethiols and 6-mercapto-1-hexanol show notably high correlation.

Figure 1: Inter-class separability of standard Raman spectra. Both (a) the separation in PCA and (b) the low off-diagonal correlations indicate that most chemical classes are readily distinguishable. An exception is the structurally related thiols and 6-mercapto-1-hexanol, which are extremely similar. The near point-like clusters in (a) suggest minimal within-class variance in the first two principal components for this dataset.

Prior to peak extraction, reference library building and similarity scoring, the following preprocessing steps were done on all the data, including the reference data and query data (testing and validation sets).

Background subtraction is done via Asymmetric Least Squares (AsLS). The behaviour of this algorithm is controlled by two parameters: λ, which dictates the smoothness of the estimated baseline, and p, which controls its asymmetry, giving more weight to points below the estimated baseline to avoid fitting to the peaks themselves.

Following background subtraction, each spectrum is normalized using L2-normalization.

3.1.2 Characteristic Peak Extraction (CaPE)

The first stage involved Characteristc Peak Extraction (CaPE), used to build the CP library from a set of known reference spectra.

After preprocessing, the peak-finding algorithm (CaPE) is applied to every reference spectrum. CaPE works as follows: the spectra are first smoothed using a moving average filter, with the window size K determining the degree of smoothing. Peaks are then identified based on a minimum height threshold and prominence, which measures how much a peak stands out from the surrounding signal.

Peaks are only considered as a CP if it appears consistently and frequently across the multiple reference spectra for a specific analyte. The algorithm selects the N most frequent peaks for each chemical to construct the CP library-a refined dataset that maps each chemical to its unique set of characteristic peak wavenumbers. This set of peaks serves as the chemicals unique spectral fingerprint.

3.1.3 Similarity Scoring

The second stage is the classification via similarity scoring. When a query spectrum is introduced, it first undergoes the same preprocessing steps. Then, for each potential match, a query feature vector is constructed. This is a critical step: the vector is created by extracting the intensity values from the query spectrum at the wavenumber locations corresponding to the characteristic peaks of the chemical being tested. To account for minor peak shifts between the query and reference spectra, the intensity is not taken from a single point but is instead the maximum value within a small window of width w centered on the characteristic peak's location. This feature vector, which represents how well the query spectrum expresses the fingerprint of the reference chemical, is then compared to a corresponding set of reference feature vectors derived from the original reference spectra. The degree of similarity is quantified using the dot product between the query's feature vector and the reference feature vectors, yielding a final CaPSim score. The query spectrum is ultimately assigned the identity of the reference chemical that produces the highest score.

The performance of the CaPSim algorithm is highly dependent on the set of hyper parameters (λ, p, K, *height*, *prominence*, N and w). To ensure optimal performance, their values were not chosen arbitrarily or even in lock step with what is reported in [1], but were determined through a grid search analysis, where a wide range of parameter combinations was evaluated to find the set that maximized the overall classification accuracy.

Their values are summarized in the following Table 5:

Hyperparameter	**Value**
λ	1×10^4
p	0.01
K	3
N	12
w	15
height	0.01
prominence	None

Table 5: Hyperparameter values for the CaPSim Algorithm

3.1.4 CaPSim Performance

To quantitatively assess the performance of the implemented CaPSim algorithm, a validation was conducted using the standard Raman dataset. The dataset was partitioned such that half of the spectra were used to construct the reference CP Library, while the remaining half served as a set of unknown query spectra. This approach simulates a realistic scenario where the model must identify spectra it has not previously seen. The algorithm's ability to correctly rank the true chemical identity for each query spectrum is summarized in Table 6. Performance is reported in terms of Top-K accuracy, which evaluates the presence of the correct label within the top-ranked results.

Top-K	Accuracy (%)
1	30.04
2	85.16
3	85.16
4	96.47
5	100.00

Table 6: CaPSim Top-K classification accuracy on the standard Raman test set. Top-K accuracy represents the percentage of query spectra for which the correct chemical identity was found within the K most similar results as ranked by the algorithm. For instance, Top-1 accuracy (30.04%) indicates the cases where the single most similar reference was the correct one, while Top-5 accuracy (100.00%) indicates that the correct label was always present within the five highest-ranked predictions.

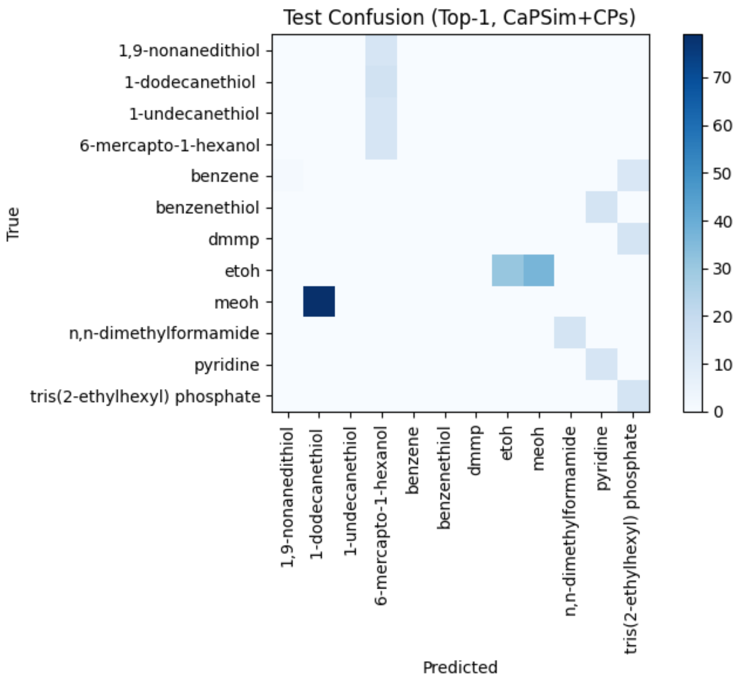


Figure 2: Top-1 confusion matrix on the held-out test split using CaPSim with characteristic-peak features (CPs). Cells show counts of spectra with true class (rows) predicted as (columns), with darker shades indicating higher counts. Notable errors concentrate among the structurally related alkanethiols and 6-mercapto-1-hexanol, and between the alcohols (ethanol/methanol).

3.2 Enhancing CaPSim with a k-Nearest Neighbours (kNN) Classifier

The CaPSim method provided an elegant method for ranking potential matches that was grounded in the physical nature of the data. However, as seen in the previous Section 3.1.4 and shown in Table 6 and Figure 2, its performance as a classifier was limited, achieving only $\sim 30\%$ Top-1 accuracy. Given that the Top-2,3,4,5 accuracies were $> 85\%$, this suggested that the feature extraction approach was sound but that the final classification step required a more sophisticated model, especially in the case of extremely similar spectra. Therefore, the next step was to integrate the feature extraction from CaPSim (which was CaPE) with a k-Nearest Neighbours (kNN) classifier.

The kNN algorithm is a simple, non-parametric method that classifies a new data point based on the majority class of its 'k' closest neighbors in a feature space. The core of this enhanced method involves a crucial modification to how the spectral features are defined. Instead of creating different feature vectors for a query spectrum against each reference class, a single, global set of characteristic peaks is first established by taking the union of all characteristic peaks from all classes in the training data. This creates a unified and consistent feature space where every spectrum is represented by a feature vector of the same length.

The workflow for this method involves partitioning the reference dataset itself. Half of the reference spectra are designated as a training set, while the other half are held out as a test set. The training set is used to establish the global set of characteristic peaks and to train the kNN model. Each spectrum in the training set is transformed into a feature vector based on the intensities at these global peak locations. The kNN model learns the relationship between these feature vectors and their corresponding chemical labels. To classify a new, unknown query spectrum, it is first converted into a feature vector using the same global set of peaks. This vector is then passed to the trained kNN model, which identifies the 'k' most similar training vectors (its "nearest neighbors") using a distance metric (in this case, cosine distance) and assigns the query the chemical label that is most common among those neighbors. This two-step process formalizes the classification, leveraging a learned decision-making process within a consistent feature space and leading to a more robust framework with improved accuracy.

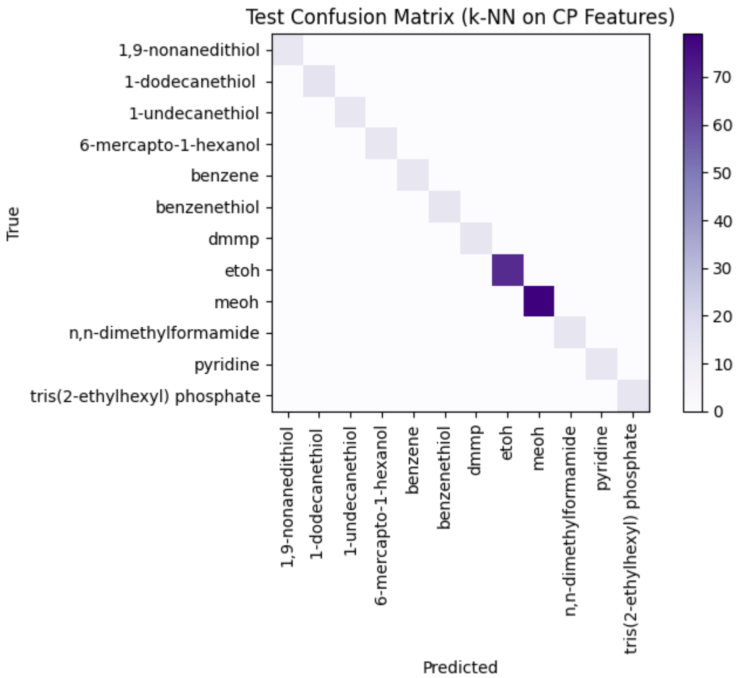


Figure 3: Confusion matrix for the Top-1 classification performance of the CaPSim-kNN model on the validation query set. The rows represent the true chemical labels, and the columns represent the predicted labels. The matrix shows perfect classification accuracy, with all predictions lying on the main diagonal. This indicates that for every query spectrum, the model correctly identified the chemical, resulting in zero misclassifications across all classes in the set.

3.2.1 CaPSim-kNN Performance

As illustrated in Figure 3, the performance of the CaPSim-kNN model was flawless, achieving 100% Top-1 accuracy on the test set. The confusion matrix visually confirms that every query spectrum was correctly identified, with no misclassifications. This perfect result demonstrates the significant improvement gained by pairing the CaPE feature extraction with a formal kNN classification model.

4 Advancing to Deep Learning: Siamese Networks

The success of the CaPE-kNN approach highlighted the importance of having a good feature representation of the spectra. While CaPE/CaPSim provides a rules-based method for feature extraction, the next logical step was to explore if a model could *learn* an optimal feature representation directly from the data.

Furthermore, the facial recognition analogy used in [1] to frame the CaPSim problem is, in fact, a classic use case for a specific deep learning architecture: Siamese Networks. These models are explicitly designed to learn a mapping from a high-dimensional input (like an image or spectrum) to a lower-dimensional embedding space where the distance between two points corresponds to their similarity, making them ideal for this type of identification task.

4.1 A Deep Learning Primer

A neural network is a stack of linear layers interleaved with nonlinear activations that learns a mapping from input x to output y by adjusting weights to minimize a loss function. A 1D convolutional neural network (CNN) slides small filters along the spectrum to detect local patterns (e.g., peaks), composing them into higher-level features across layers. An *embedding* is a low-dimensional numeric representation that places similar spectra near each other in a learned space. Training proceeds by gradient descent: the loss function measures "how wrong" the current predictions are, and back-propagation updates the weights. In *contrastive learning*, pairs of spectra are labeled "same chemical" or "different": the loss pulls same-class embeddings together and pushes different-class embeddings apart up to a margin, yielding a space where Euclidean distance encodes chemical similarity.

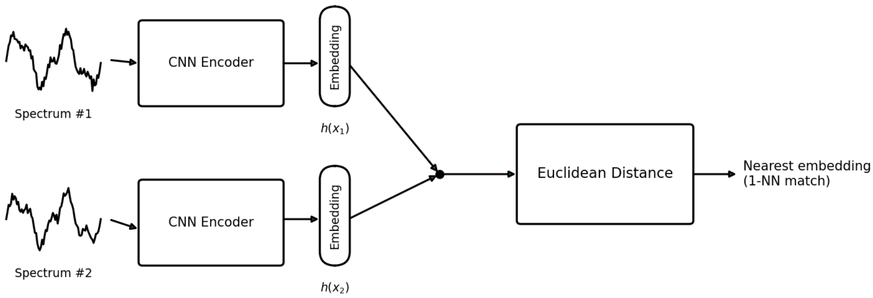


Figure 4: Siamese network concept for Raman spectra: two tied Convolutional Neural Network encoders map a query and a reference spectrum to embeddings; similarity is measured by Euclidean distance in embedding space, and the nearest reference embedding (1-Nearest Neighbour) determines the predicted label.

4.2 Siamese Networks: an Overview

A Siamese Network is a class of neural network architecture that contains two identical sub-networks that share the same parameters and weights. Unlike traditional classification networks that learn to map an input to a specific class label, a Siamese network learns to assess the similarity between two inputs. It takes a pair of data points (e.g. two Raman spectra) and outputs a similarity score, effectively converting a multi-class classification problem into a distance-based similarity problem.

In essence, a Siamese Network learns the optimal latent space representation for the given data.

The training process is what makes this architecture unique. The network is fed pairs of spectra, which can be either a "positive pair" (both spectra are from the same chemical class) or a "negative pair" (the spectra are from different classes). The network's objective, governed by a specialized loss function like contrastive loss (Equation 1), is to generate low-dimensional feature vectors, or "embeddings," for each spectrum. For positive pairs, the network is trained to minimize the distance (e.g., Euclidean distance) between their embeddings. For negative pairs, it is trained to maximize the distance between their embeddings, but only up to a certain predefined margin. Over many iterations, this process forces the network to learn an embedding space where similar items are geometrically close to each other and dissimilar items are far apart.

$$L\Big(W,\ (Y,\vec{X}_1,\vec{X}_2)^i\Big) = (1-Y)\, L_S\Big(D_W^i\Big) + Y\, L_D\Big(D_W^i\Big)\,. \tag{1}$$

This approach has several advantages that make it particularly well-suited for spectroscopic applications. In [2], Siamese networks were shown to excel in "few-shot" learning scenarios where training data is limited or unbalanced.

4.3 Standard Raman Classification

The Siamese Network approach was first validated on the standard Raman single spectra dataset, employing a data strategy that highlights the model's strength in few-shot learning scenarios. The experiment was structured using three distinct datasets:

1. **Training Set**: Containing five spectra per chemical class was used exclusively to train the Siamese network's encoder. The network learned to create an optimal embedding space by comparing pairs of these spectra using a contrastive loss function.

2. **One-Shot Reference Set**: Containing only a single spectrum for each chemical, was used to establish the known locations of each class within the learned embedding space.

3. **Test Set**: The dataset from Section 2.2.1 was used to evaluate the model's performance.

The classification process involved first passing the one-shot reference set and the query set through the trained encoder to generate their respective embeddings. Then, for each query spectrum's embedding, the nearest neighbor was found within the reference embeddings using Euclidean distance. The query was assigned the label of this nearest neighbor.

4.3.1 Siamese Network Performance on Standard Raman

This methodology demonstrates the power of the Siamese architecture: after learning a meaningful representation of similarity, it can perform accurate classification using only a single reference example per class, as shown in Figure 5, a significant advantage over traditional models that require extensive training data for every class.

4.4 One-Shot Learning with Siamese Networks for SERS

Having demonstrated its effectiveness on standard Raman data, the Siamese Network was next applied to the more challenging task of SERS classification. The primary

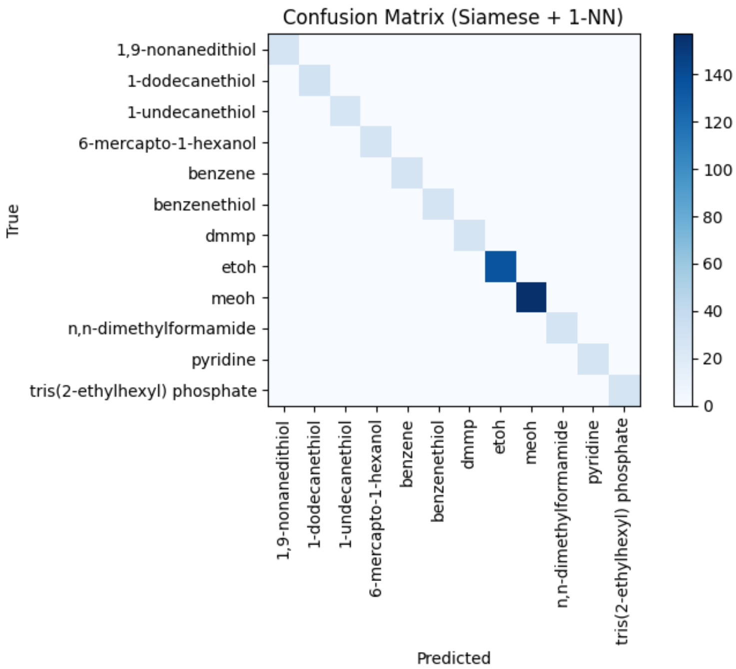


Figure 5: Confusion matrix for the Siamese-embedding + 1-NN classifier on standard Raman single spectra. All samples fall on the diagonal (perfect Top-1 accuracy): the nearest embedding in the learned space is always the correct chemical.

difficulty with SERS data is the significant spectral variability introduced by different substrates. To address this, the classification target was redefined: instead of identifying only the chemical, the model was tasked with identifying the unique chemical-substrate combination.

The data handling strategy was designed to test the model's performance in a realistic, few-shot learning environment. The full SERS dataset was partitioned such that only 20% of the spectra for each class were used for training the Siamese network's encoder. The remaining 80% of the data was held out as a query set for evaluation. A separate reference set was created by computing the average spectrum for each unique chemical-substrate pair. This provides a stable, prototypical representation for each class in the reference library.

The classification workflow mirrored the previous experiment: the trained encoder was used to generate embeddings for the averaged reference set and the query set. Each query spectrum was then classified by finding the nearest neighbor in the reference set's embedding space. This approach proved to be highly effective, achieving a Top-1 accuracy of 98.76%. The detailed performance is shown in the confusion matrix in Figure 6. The model demonstrates near-perfect accuracy, with only a few misclassifications, underscoring the robustness of using a learned embedding space to overcome the challenges of SERS spectral variability.

4.5 Classifying Raman Mixtures with a Siamese Network and Multi-Layer Perceptron (MLP)

The project next addressed the classification of two-component standard Raman mixtures, a task that presents a significant increase in complexity due to the overlapping and convoluted spectral features, where often one component's signal completely dominates the other. To tackle this, a two-stage hybrid model was developed, combining the feature-learning power of a Siamese Network with the classification capabilities of a Multi-Layer Perceptron.

An MLP is a functional type of Feed-Forward Neural Network (FNN), consisting of an input layer, one or more "hidden" layers of neurons, and an output layer. Each neuron receives inputs from the previous layer, applies a weighted sum, and passes the result through a non-linear activation function. By adjusting these weights during training, an MLP can learn to approximate highly complex, non-linear relationships between its inputs and outputs.

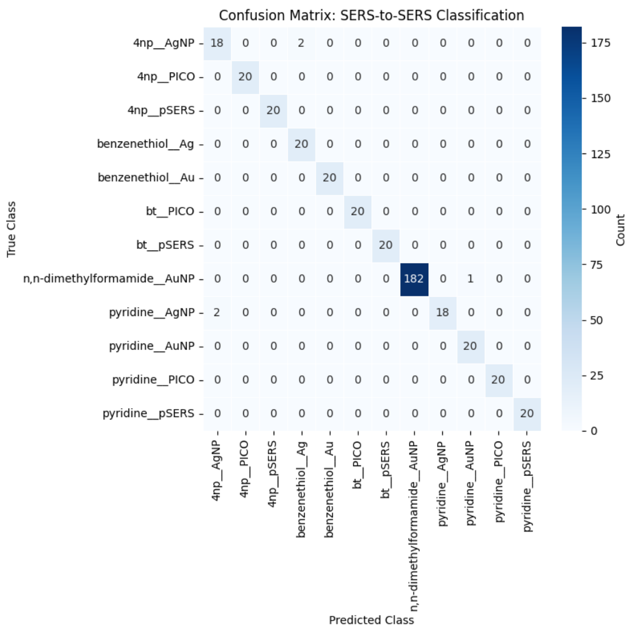


Figure 6: Confusion matrix for SERS-to-SERS classification using Siamese embeddings with 1-NN (rows: true, columns: predicted). Performance is near perfect; the few errors are substrate-consistent. This pattern suggests the embedding preserves substrate effects even when the chemical is confused.

4.5.1 Siamese + MLP Workflow

The workflow was structured as follows: First, a large dataset of synthetic mixtures was generated by taking all possible pairs of pure chemical spectra from the reference library (dataset in 2.2.1) and linearly combining them across a range of concentration ratios, with a small amount of random noise added to simulate experimental variance. A Siamese Network was then trained on this synthetic dataset. Its goal was not to identify the individual components, but to learn a rich embedding space that could effectively represent the complex features of the mixed spectra.

Once the Siamese network was trained, its encoder was used as a fixed feature extractor. The entire synthetic mixture dataset was passed through this encoder to generate a corresponding dataset of low-dimensional embeddings. These embeddings then served as the input for the MLP. The MLP was trained to perform a multi-label classification task: for each input embedding, its output layer predicts the presence or absence of every chemical in the library.

4.5.2 Siamese + MLP Performance

Chemical	Precision	Recall	F1-Score	Support
1-dodecanethiol	0.84	0.77	0.8	243
6-mercapto-1-hexanol	0.88	0.33	0.48	108
Benzene	1.00	1.00	1.00	193
Benzenethiol	0.67	1.00	0.8	72
Ethanol	1.00	1.00	1.00	121
Methanol	1.00	0.96	0.98	243
n,n-dimethylformamide	1.00	1.00	1.00	72
Pyridine	1.00	1.00	1.00	108
Macro Average	0.93	0.88	0.91	1160

Table 7: Per-class precision, recall, and F1 for the Siamese-embedding + MLP classifier on the *real mixtures* dataset. Each mixture is multi-labeled by its constituent chemicals; **Support** is the number of mixtures in which a chemical appears (total label-occurrences = 1160). Most classes are near-perfect (Benzene, Ethanol, n,n-dimethylformamide, Pyridine); Methanol shows slightly reduced recall, while the long-chain thiols are hardest—1-dodecanethiol has moderate recall and 6-mercapto-1-hexanol low recall (0.33), consistent with their spectral similarity. **Precision** measures correctness of positive predictions $= \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}$; **Recall** measures completeness $= \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}$; **F1** is their harmonic mean $= \frac{2PR}{P+R}$. Overall averages are high (precision ≈ 0.93, recall ≈ 0.88, F1 ≈ 0.91).

The results displayed in Table 7 on the real mixtures test set demonstrate the model's strong overall performance, achieving an average F1-score of 0.91. While the model perfectly identified several components, its primary challenge lay in distinguishing between spectrally similar thiols. This is most evident in the low recall for *6-mercapto-1-hexanol*, indicating that while the model rarely makes incorrect positive predictions, it frequently fails to identify this component when it is present, a difficulty consistent with the high spectral overlap among the thiol compounds.

5 A Clustering Analysis

The remarkable success of the Siamese Network models raised a critical question: did the networks truly learn a more optimal, geometrically structured representation of the spectral data? To answer this, a clustering analysis was performed to viusally and quantitavely compare the structure of the raw spectral data against the structure of the learned Siamese Embeddings.

5.1 Clustering Techniques & Metrics

This analysis employed a suite of techniques to probe the data's structure. For visualization, the high-dimensional spectral data was projected into two dimensions using PCA, and two non-linear manifold techniques. The first, t-Distributed Stochastic Neighbour Embedding (t-SNE), is used to reveal local cluster structure by arranging points in a 2D map that preserves the neighbourhood probabilities of the original data. The second, Uniform Manifold Approximation and Projection (UMAP), is a more modern technique that is often faster and better at preserving the global structure of the data, meaning the relative positions of clusters are more meaningful.

For quantitative evaluation, the HDBSCAN algorithm was used. This powerful density-based method does not require a pre-specified number of clusters and can identify clusters of varying shapes while designating sparse points as noise. The quality of the resulting clusters was measured against the true chemical labels using several standard metrics:

1. **Adjusted Rand Index (ARI)**: Measures the similarity between the ground truth labels and the predicted clsutering results. A score of 1.0 indicates a perfect match.

2. **Purity**: Each cluster is assigned the label of the most frequent true class within it. It is the fraction of the total samples that were correctly assigned.

3. **Inter/Intra-class Distance Ratio**: This metric provides a powerful measure of class separability. It is the ratio of the average distance between samples of different classes (inter-class distance) to the average distance between samples of the same class (intra-class distance). A higher ratio signifies that the classes are both internally compact and well-separated from each other.

5.2 Clustering Results

For the standard Raman data, the analysis confirmed that the classes were already well separated in the raw spectral space, and both the raw and the Siamese embeddings yielded a perfect clustering score (ARI = 1.0), as seen in Figure 7.

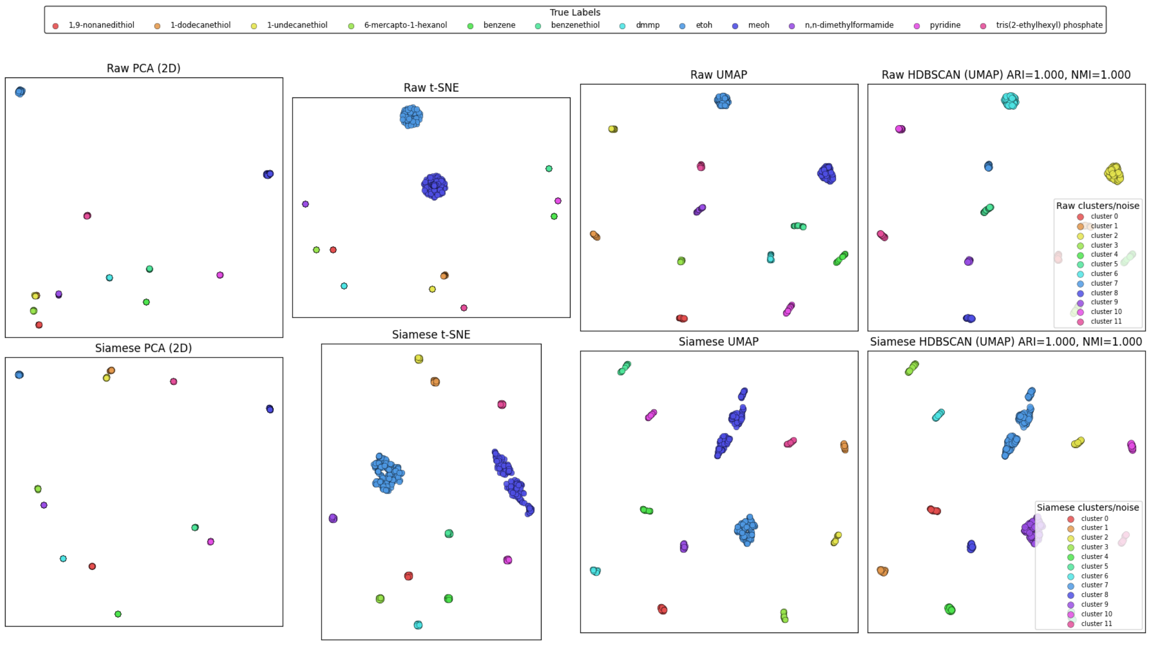


Figure 7: Comparison of dimensionality reduction and clustering on raw standard Raman spectra (top row) versus their learned Siamese embeddings (bottom row). Both representations show perfectly separated classes, confirmed by the flawless ARI score of 1.0, indicating that the standard Raman data is inherently well-structured.

However, the analysis of the SERS data revealed the true power of the learned representation. As shown in Figure 8, the 2D visualizations of the raw SERS spectra show significant overlap and poor separation between classes. In stark contrast, the visualizations of the Siamese embeddings show more distinct, and well-separated clusters.

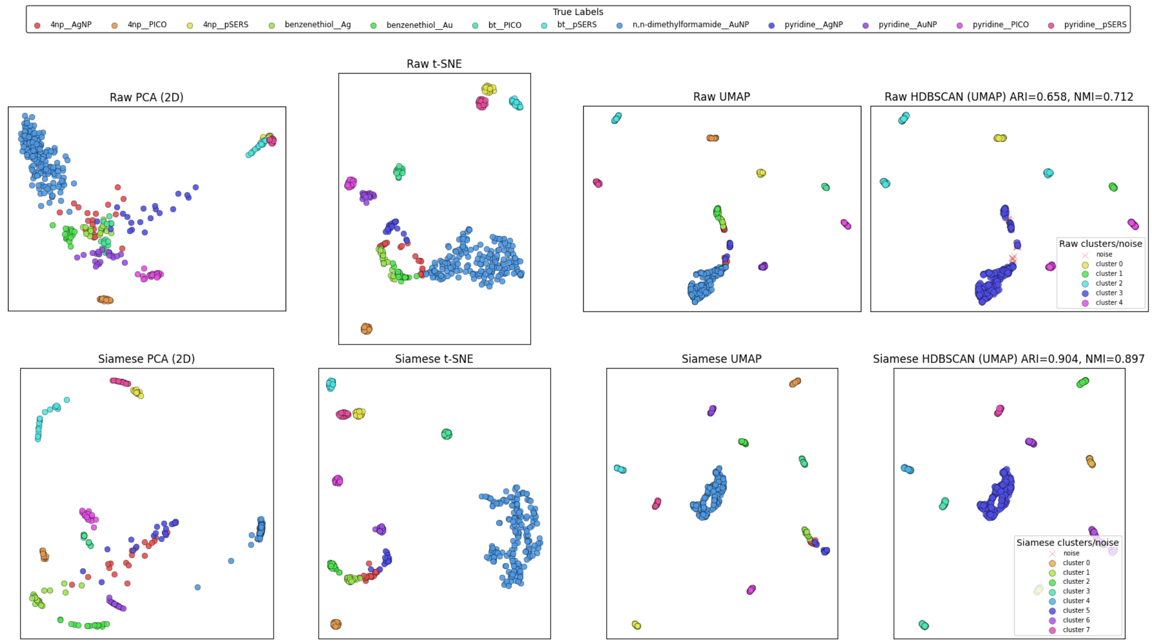


Figure 8: Comparison of dimensionality reduction and clustering on raw SERS spectra (top row) versus their learned Siamese embeddings (bottom row). A dramatic improvement in class separation is visually evident in the Siamese embedding space across all visualization methods. The embeddings form tight, distinct clusters, unlike the overlapping and poorly defined clusters of the raw data.

This visual improvement was confirmed quantitatively. HDBSCAN on the raw SERS data produced a poor match with the true labels (ARI = 0.658, Purity = 0.680). The inter/intra class distance ratio was only 2.17, indicating modest separation.

The clustering on the Siamese embeddings, however, was substantially better (ARI = 0.9, Purity = 0.8), and the inter-intra class distance ratio increased dramatically to 15.20; a seven fold improvement. This improvement in the separability ratio provides compelling evidence that the Siamese Networks successfully learned the most optimal representation of the spectral data.

Figure 9 below, again showing that the clustering errors are substrate consistent with 4np_AgNP, Benzenethiol_Ag and Pyridine_AgNP being clustered together. Interestingly Benzenethiol_Au is also clustered with the previously mentioned chemical-substrate pairs.

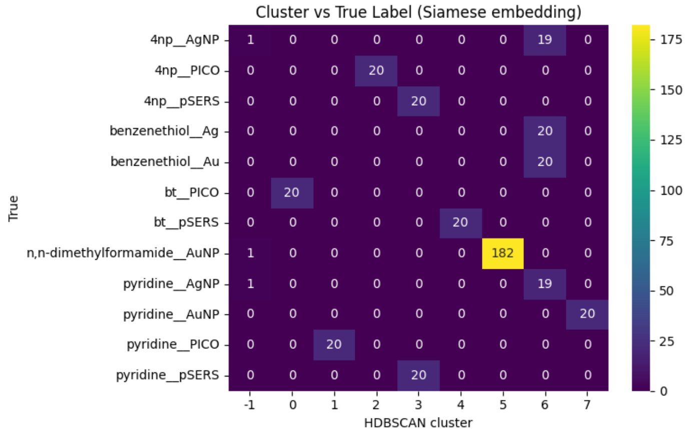
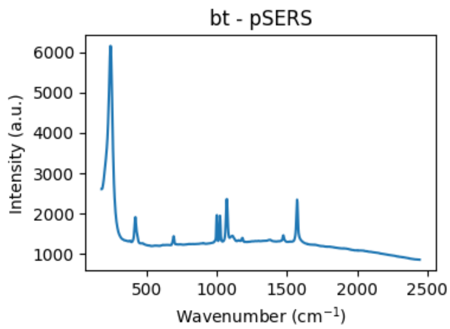


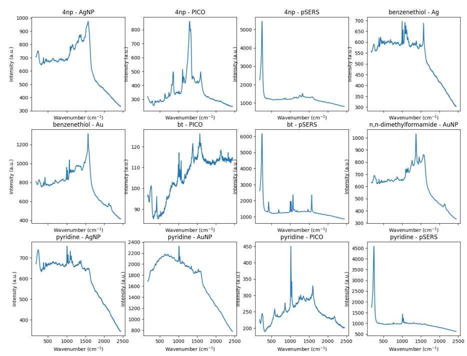
Figure 9: HDBSCAN clustering in the Siamese-embedding space (rows: true chemical–substrate labels; columns: discovered clusters, -1 = noise). Most classes map cleanly to single clusters. Notably, cluster 6 groups four chemical–substrate pairs, three of which share the same silver-based substrate (Ag/AgNP), suggesting substrate-driven structure in the embedding.

5.2.1 Follow-up: Impact of of Instrumental Artifact Removal

During the initial analysis, a consistent instrumental artifact was observed in the SERS data around 300cm^{-1} as shown below in Figure 10.



(a) The average Benzethiol with pSERS sensor zoomed in.



(b) The average spectrum for each SERS chemical-substrate pair in the SERS dataset.

Figure 10: Average spectrum for the SERS data. A distinctive, and in some cases, dominant peak at around 300cm^{-1} can be observed.

To Investigate is impact on the model's performance, a follow-up clustering analysis was conducted on a modified dataset where all spectral data below 330cmcm^{-1} were removed.

Interestingly, the HDBSCAN results on this cleaned data, shown in Figure 11, revealed a new clustering pattern, While most chemical-substrate pairs were perfectly seperated, two distinct groups emerged. The first grouped Pyridine_AgNP and Pyridine_AuNP together and the second grouped Benzenethiol_Au, Benzenethiol_Ag and 4np_AgNP together. This suggests that for Pyridine and Benzenethiol, the chemical signature in embedding space was stronger than the substrate-induced variations. In the case of 4np_AgNP, it was shown that it was very spectrally similar to Benzenethiol (AgNP & AuNP), suggesting that for loosely-bound analytes (such as 4np), the spectra features are more heavily influenced by the substrate material than by the specific chemical identity.

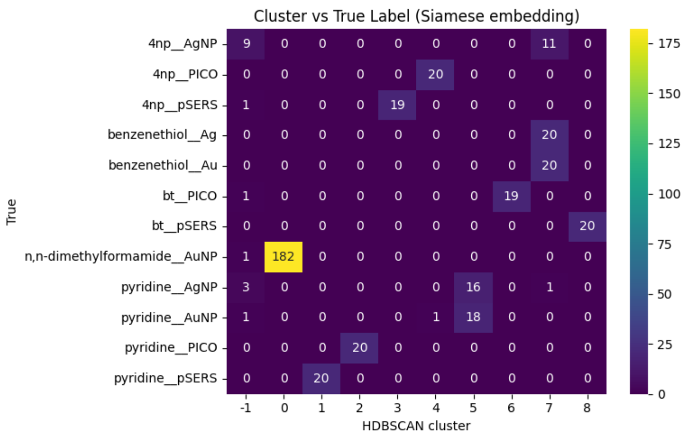


Figure 11: HDSBCAN clustering in the Siamese-embedding space on the modified dataset with spectral data below 330cm^{-1} removed.

However, a final classification using the trained Siamese Network on this truncated dataset revealed a further nuance. As shown in the confusion matrix in Figure 12, the model performed nearly perfectly, only showing minor confusion between the two pyridine samples and the two Benzenethiol samples. Critically, unlike the unsupervised HDBSCAN clustering, the supervised 1-NN classifier was able to perfectly differentiate 4np_AgNP from the Benzenethiol cluster.

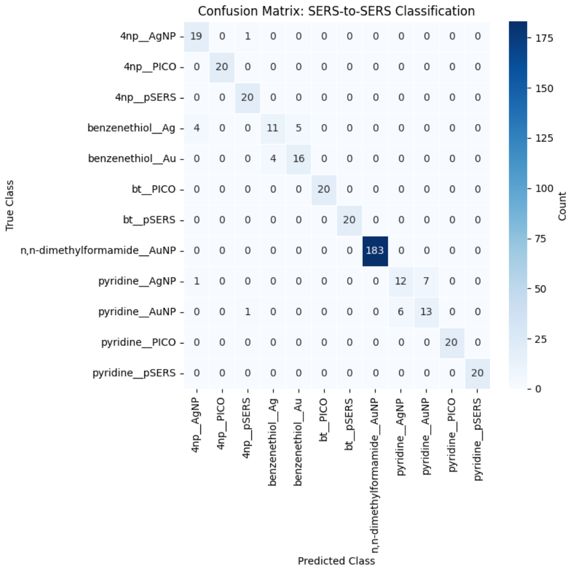


Figure 12: Confusion Matrix for SERS classification using Siamese embeddings with 1-NN on the truncated dataset.

These combined findings highlight the sophisticated nature of the learned embedding space. While unsupervised clustering reveals the dominant geometric structure (driven by chemical or substrate similarity), the supervised nearest-neighbor classification can still resolve finer distinctions. This suggests that future improvements in SERS-based classification may depend not only on model architecture but also on optimizing the SERS sensors themselves to produce more distinct signals, particularly for weakly interacting analytes.

6 Future Work

This project has successfully established a robust framework for spectral classification, and this work opens up several avenues for future research. A primary goal is to extend these models beyond simple identification towards quantification, particularly for the analysis of chemical mixtures. Developing a model that can accurately predict the relative concentrations of components from a single mixed spectrum would be a significant advancement.

Additionally, while the Siamese+MLP model performed well on mixtures, its difficulty in resolving spectrally similar thiols highlights an area for improvement. Future work could explore more advanced neural network architectures or training strategies specifically designed to enhance the model's ability to distinguish between these challenging components. Finally, the insights gained from the SERS clustering analysis—particularly the interplay between chemical and substrate signatures—suggest that further progress could be made by co-developing SERS substrates and machine learning models in tandem to produce more distinct and classifiable signals.

A Experimental Details and Hyperparameters

A.1 General Preprocessing

Parameter	Value / Description
Baseline correction	Asymmetric Least Squares (AsLS)
AsLS λ (smoothness)	1×10^4
AsLS p (asymmetry)	0.01
AsLS iterations	10
Normalization	l_2-norm

Table 8: General preprocessing parameters. (Source: Implemented in all scripts, e.g., `CaPSim.py`)

A.2 CaPSim/CaPE/kNN

Parameter	Value / Description
Data split (CaPSim)	Standard Raman: 50% reference library / 50% query
Data split (kNN)	Standard Raman: 50% train / 50% test (from reference data)
Characteristic Peak Extraction (CaPE)	
Smoothing window K	3
Peak height threshold	0.01
Number of peaks N	12
Peak window width w	15
k-Nearest Neighbors	
Neighbors k	3
Distance metric	Cosine

Table 9: CaPSim, CaPE, and kNN settings. (Source: `CaPSim.py`, `CaPSim_kNN.py`)

A.3 Siamese Network

Parameter	Value / Description
Encoder Architecture	
Layer 1	Conv1D (16 filters, kernel=7) + ReLU + MaxPool(2)
Layer 2	Conv1D (32 filters, kernel=5) + ReLU + MaxPool(2)
Dense / embedding	Linear → 64
Training	
Loss	Contrastive (margin = 1.0)
Optimizer	Adam; learning rate 1×10^{-3}
Epochs	100 (Standard & SERS), 10 (Mixtures)
Batch size	32
Augmentation	Gaussian noise ($\sigma = 0.01$) + random shift (± 2)
Data Splits	
Standard Raman	Train: 5 spectra/class; reference: 1 spectrum/class
SERS	Train: 20% per class; reference: averaged spectrum/class
Mixtures	Trained on synthetic mixtures from all pure components

Table 10: Siamese network encoder and training details. (Source: `Siamese_Network_OneShot.ipynb`, `Siamese_MLP_3.ipynb`)

A.4 MLP for mixtures

Parameter	**Value / Description**
Data split	Synthetic embeddings: 80% train / 10% val / 10% test
Architecture	
Input	64 (from Siamese embedding)
Hidden	128 (ReLU) $\rightarrow$ 64 (ReLU)
Output	Linear (number of classes); sigmoid at inference
Training	
Loss	BCEWithLogitsLoss (positive class weighting)
Optimizer	Adam; learning rate 1×10^{-3}
Epochs	200
Prediction threshold	0.5

Table 11: MLP architecture and training for mixture classification. (Source: `Siamese_MLP_3.ipynb`)

References

[1] Yilong Ju, Oara Neumann, Mary Bajomo, Yiping Zhao, Peter Nordlander, Naomi J. Halas, and Ankit Patel. Identifying surface-enhanced raman spectra with a raman library using machine learning. *ACS Nano*, 17:21251–21261, 2023. doi: 10.1021/acsnano.3c05510.

[2] Jhonatan Contreras, Sara Mostafapour, Jürgen Popp, and Thomas Bocklitz. Siamese networks for clinically relevant bacteria classification based on raman spectroscopy. *Molecules*, 29(5):1061, 2024. doi: 10.3390/molecules29051061.